Exp No: 5
10/8/26

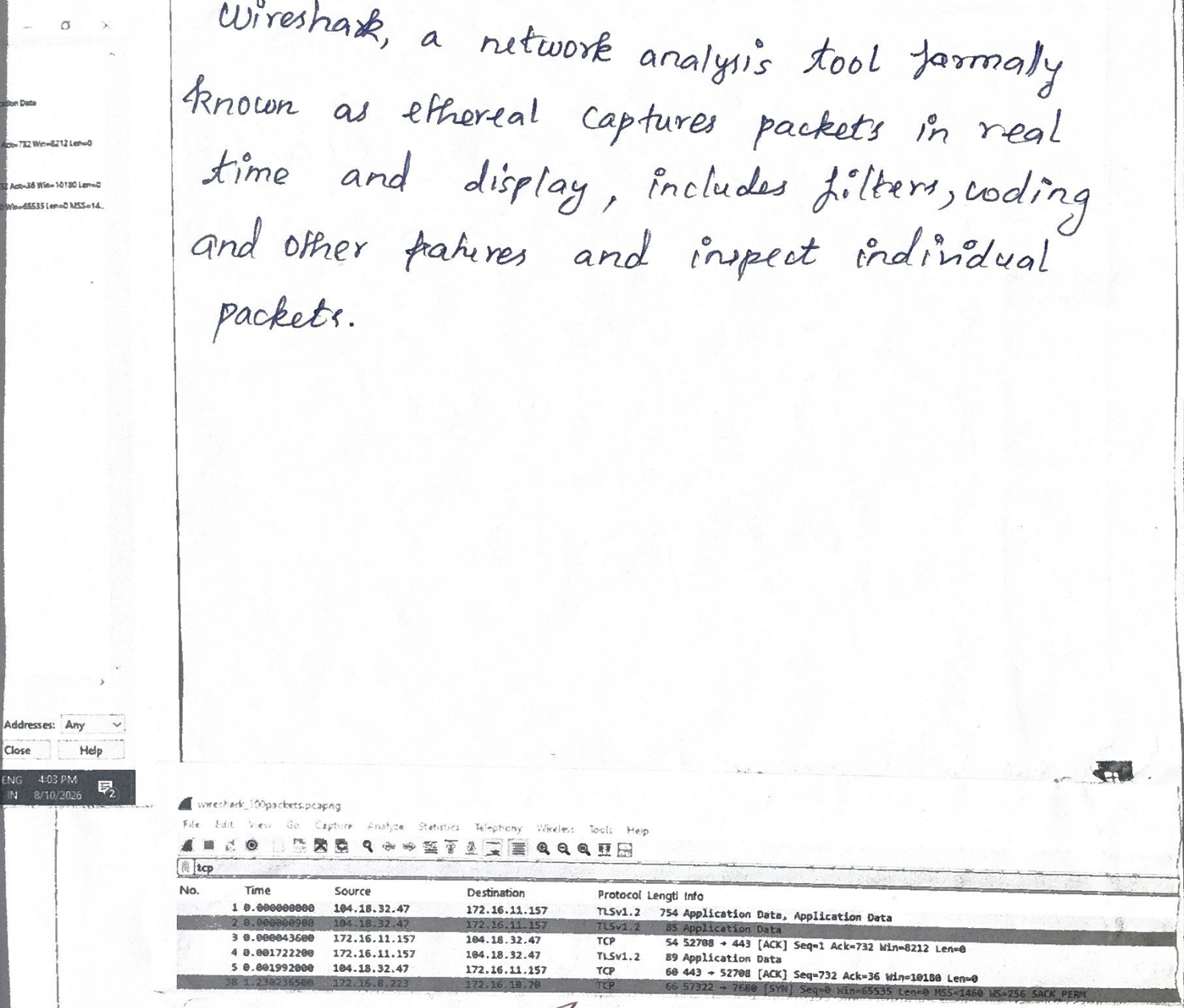
EXPERIMENT ON PACKET CAPTURE TOOL : WIRESHARK

AIM:

To capture filter, and always analyze the network packets using wireshark packet capture tool and to study different network protocols as TCP, UDP, ARP, DNS, HTTP, packet details and flow graphs.

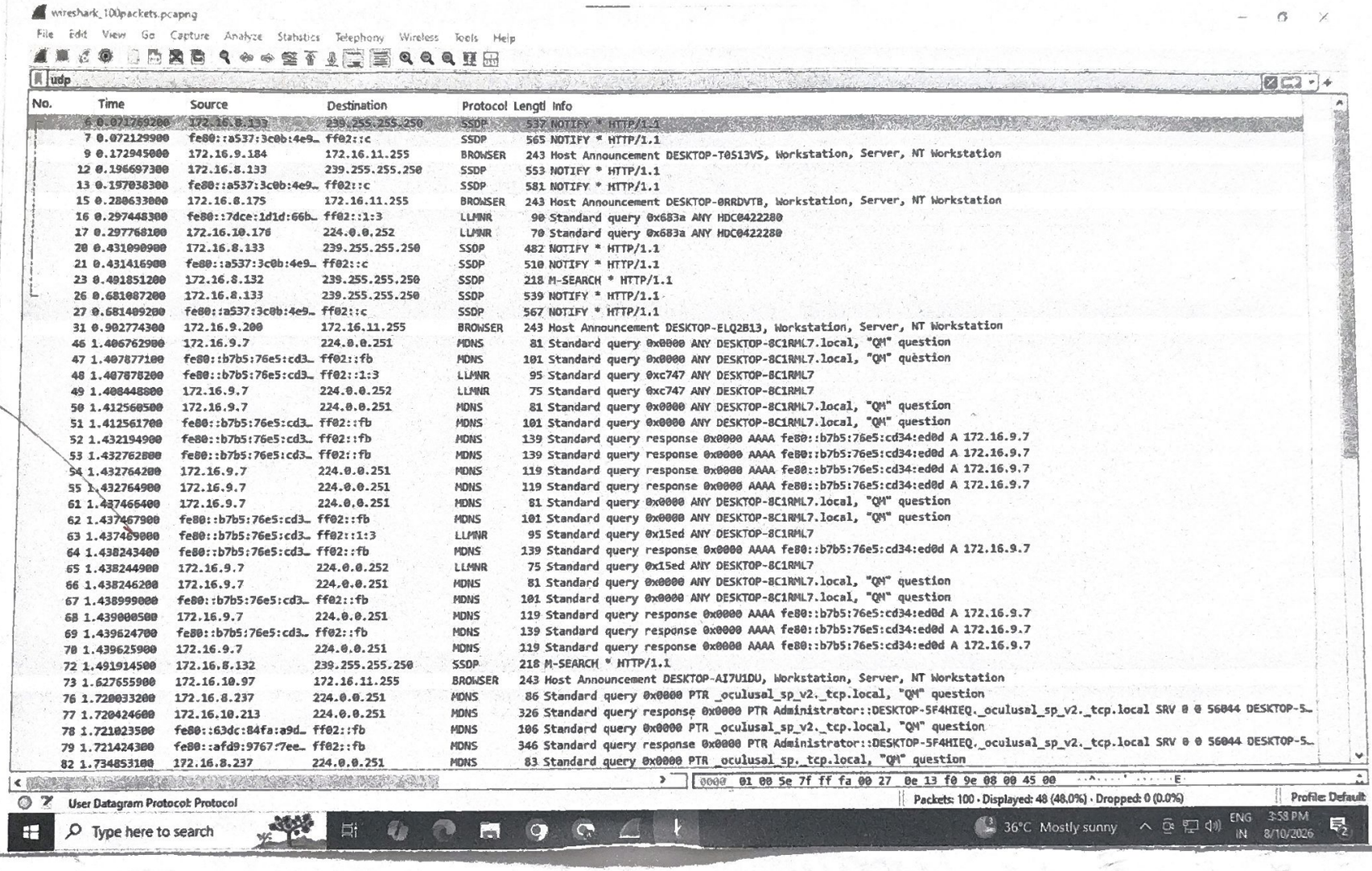
WIRESHARK:

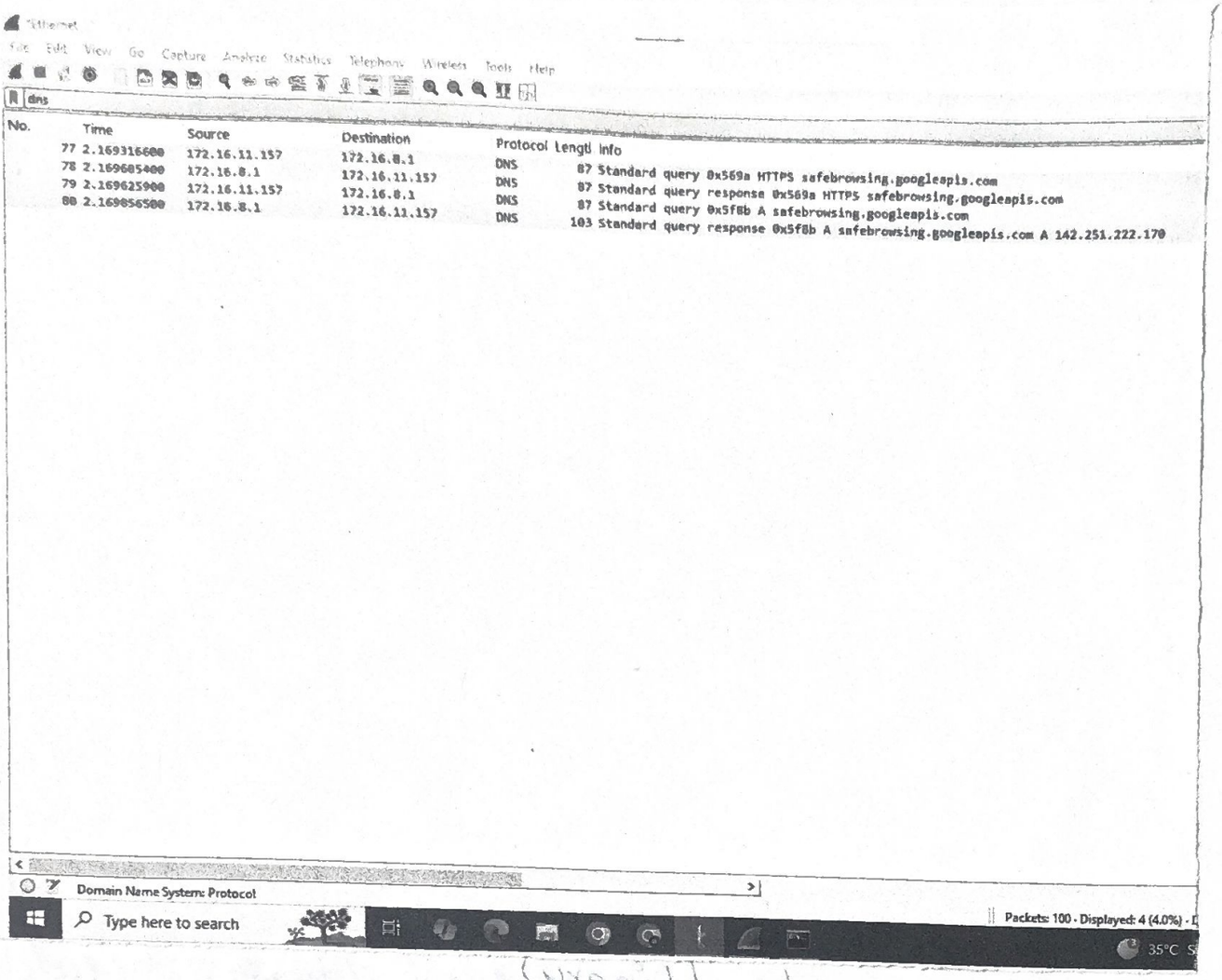
Wireshark, a network analysis tool formally known as ethereal captures packets in real time and display, includes filters, coding and other features and inspect individual packets.

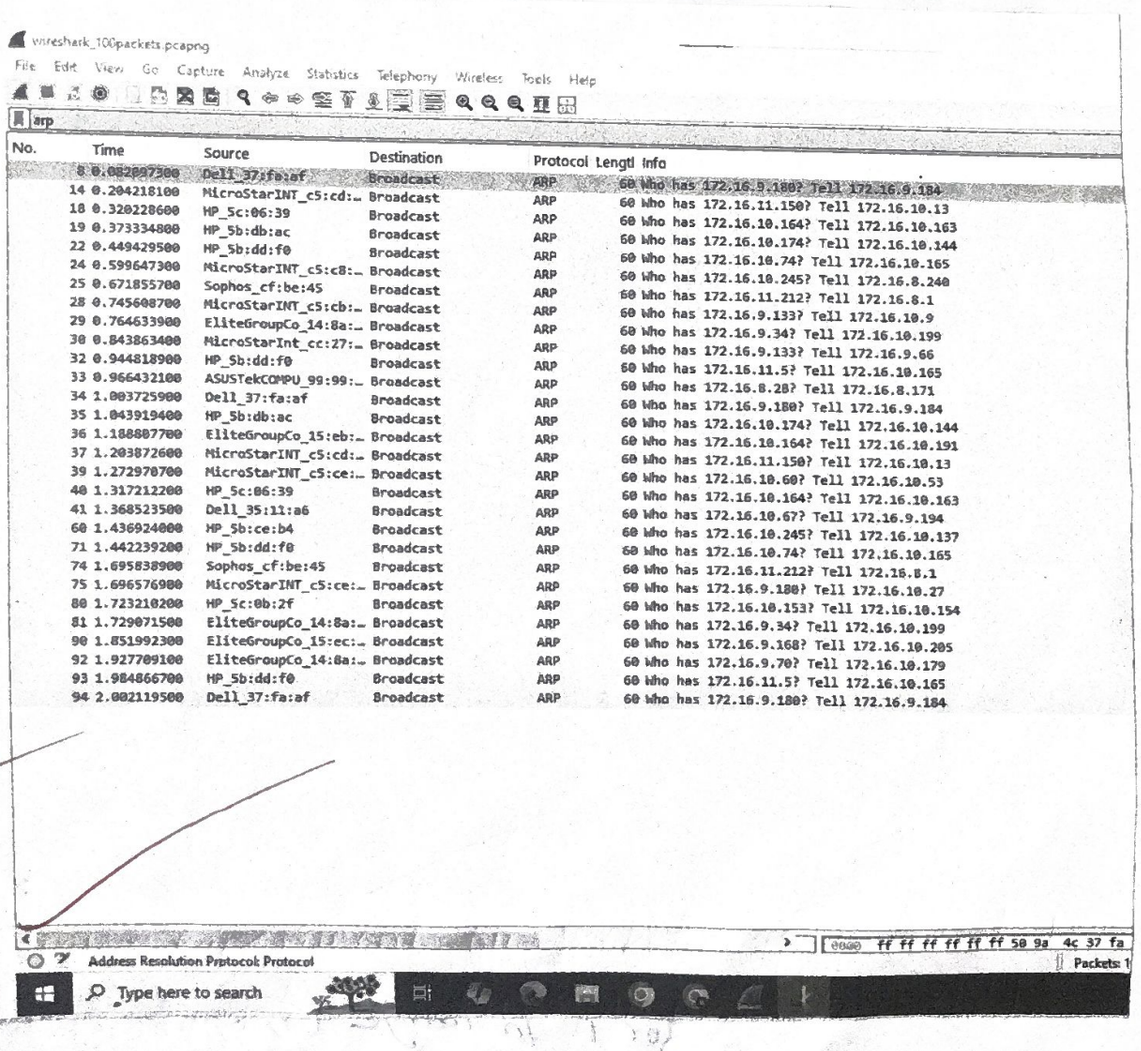


Result:

Thus the experiment on packet capture tool by using Wireshark is successfully implemented.


wireshark_100packets.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
No. Time Source Destination Protocol Length Info
6 0.071769200 172.16.8.133 239.255.255.250 SSDP 537 NOTIFY * HTTP/1.1
7 0.072129900 fe80::a537:3c0b:4e9… ff02::c SSDP 565 NOTIFY * HTTP/1.1
9 0.172945000 172.16.9.184 172.16.11.255 BROWSER 243 Host Announcement DESKTOP-T0S13VS, Workstation, Server, NT Workstation
12 0.196697300 172.16.8.133 239.255.255.250 SSDP 553 NOTIFY * HTTP/1.1
13 0.197038300 fe80::a537:3c0b:4e9… ff02::c SSDP 581 NOTIFY * HTTP/1.1
15 0.280633000 172.16.8.175 172.16.11.255 BROWSER 243 Host Announcement DESKTOP-0RRDVTB, Workstation, Server, NT Workstation
16 0.297448300 fe80::7dce:1d1d:66b… ff02::1:3 LLMNR 90 Standard query 0x683a ANY HDC0422280
17 0.297768100 172.16.10.176 224.0.0.252 LLMNR 70 Standard query 0x683a ANY HDC0422280
20 0.431090900 172.16.8.133 239.255.255.250 SSDP 482 NOTIFY * HTTP/1.1
21 0.431416900 fe80::a537:3c0b:4e9… ff02::c SSDP 510 NOTIFY * HTTP/1.1
23 0.491851200 172.16.8.132 239.255.255.250 SSDP 218 M-SEARCH * HTTP/1.1
26 0.681087200 172.16.8.133 239.255.255.250 SSDP 539 NOTIFY * HTTP/1.1
27 0.681409200 fe80::a537:3c0b:4e9… ff02::c SSDP 567 NOTIFY * HTTP/1.1
31 0.902774300 172.16.9.200 172.16.11.255 BROWSER 243 Host Announcement DESKTOP-ELQ2B13, Workstation, Server, NT Workstation
46 1.406762900 172.16.9.7 224.0.0.251 MDNS 81 Standard query 0x0000 ANY DESKTOP-8C1RML7.local, "QM" question
47 1.407877100 fe80::b7b5:76e5:cd3… ff02::fb MDNS 101 Standard query 0x0000 ANY DESKTOP-8C1RML7.local, "QM" question
48 1.407878200 fe80::b7b5:76e5:cd3… ff02::1:3 LLMNR 95 Standard query 0xc747 ANY DESKTOP-8C1RML7
49 1.408448800 172.16.9.7 224.0.0.252 LLMNR 75 Standard query 0xc747 ANY DESKTOP-8C1RML7
50 1.412560500 172.16.9.7 224.0.0.251 MDNS 81 Standard query 0x0000 ANY DESKTOP-8C1RML7.local, "QM" question
51 1.412561700 fe80::b7b5:76e5:cd3… ff02::fb MDNS 101 Standard query 0x0000 ANY DESKTOP-8C1RML7.local, "QM" question
52 1.432194900 fe80::b7b5:76e5:cd3… ff02::fb MDNS 139 Standard query response 0x0000 AAAA fe80::b7b5:76e5:cd34:ed0d A 172.16.9.7
53 1.432762800 fe80::b7b5:76e5:cd3… ff02::fb MDNS 139 Standard query response 0x0000 AAAA fe80::b7b5:76e5:cd34:ed0d A 172.16.9.7
54 1.432764200 172.16.9.7 224.0.0.251 MDNS 119 Standard query response 0x0000 AAAA fe80::b7b5:76e5:cd34:ed0d A 172.16.9.7
55 1.432764900 172.16.9.7 224.0.0.251 MDNS 119 Standard query response 0x0000 AAAA fe80::b7b5:76e5:cd34:ed0d A 172.16.9.7
61 1.437466400 172.16.9.7 224.0.0.251 MDNS 81 Standard query 0x0000 ANY DESKTOP-8C1RML7.local, "QM" question
62 1.437467900 fe80::b7b5:76e5:cd3… ff02::fb MDNS 101 Standard query 0x0000 ANY DESKTOP-8C1RML7.local, "QM" question
63 1.437469000 fe80::b7b5:76e5:cd3… ff02::1:3 LLMNR 95 Standard query 0x15ed ANY DESKTOP-8C1RML7
64 1.438243400 fe80::b7b5:76e5:cd3… ff02::fb MDNS 139 Standard query response 0x0000 AAAA fe80::b7b5:76e5:cd34:ed0d A 172.16.9.7
65 1.438244900 172.16.9.7 224.0.0.252 LLMNR 75 Standard query 0x15ed ANY DESKTOP-8C1RML7
66 1.438246200 172.16.9.7 224.0.0.251 MDNS 81 Standard query 0x0000 ANY DESKTOP-8C1RML7.local, "QM" question
67 1.438999000 fe80::b7b5:76e5:cd3… ff02::fb MDNS 101 Standard query 0x0000 ANY DESKTOP-8C1RML7.local, "QM" question
68 1.439000500 172.16.9.7 224.0.0.251 MDNS 119 Standard query response 0x0000 AAAA fe80::b7b5:76e5:cd34:ed0d A 172.16.9.7
69 1.439624700 fe80::b7b5:76e5:cd3… ff02::fb MDNS 139 Standard query response 0x0000 AAAA fe80::b7b5:76e5:cd34:ed0d A 172.16.9.7
70 1.439625900 172.16.9.7 224.0.0.251 MDNS 119 Standard query response 0x0000 AAAA fe80::b7b5:76e5:cd34:ed0d A 172.16.9.7
72 1.491914500 172.16.8.132 239.255.255.250 SSDP 218 M-SEARCH * HTTP/1.1
73 1.627655900 172.16.10.97 172.16.11.255 BROWSER 243 Host Announcement DESKTOP-AI7U1DU, Workstation, Server, NT Workstation
76 1.720033200 172.16.8.237 224.0.0.251 MDNS 86 Standard query 0x0000 PTR _oculusal_sp_v2._tcp.local, "QM" question
77 1.720424600 172.16.10.213 224.0.0.251 MDNS 326 Standard query response 0x0000 PTR Administrator::DESKTOP-5F4HIEQ._oculusal_sp_v2._tcp.local SRV 0 0 56044 DESKTOP-5…
78 1.721023500 fe80::63dc:84fa:a9d… ff02::fb MDNS 106 Standard query 0x0000 PTR _oculusal_sp_v2._tcp.local, "QM" question
79 1.721424300 fe80::afd9:9767:7ee… ff02::fb MDNS 346 Standard query response 0x0000 PTR Administrator::DESKTOP-5F4HIEQ._oculusal_sp_v2._tcp.local SRV 0 0 56044 DESKTOP-5…
82 1.734853100 172.16.8.237 224.0.0.251 MDNS 83 Standard query 0x0000 PTR _oculusal_sp._tcp.local, "QM" question
0000 01 00 5e 7f ff fa 00 27 0e 13 f0 9e 08 00 45 00
User Datagram Protocol: Protocol
Packets: 100 · Displayed: 48 (48.0%) · Dropped: 0 (0.0%)
Profile: Default
Type here to search
36°C Mostly sunny
ENG IN
3:58 PM 8/10/2026


Ethernet
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
77 2.169316600 172.16.11.157 172.16.8.1 DNS 87 Standard query 0x569a HTTPS safebrowsing.googleapis.com
78 2.169605400 172.16.8.1 172.16.11.157 DNS 87 Standard query response 0x569a HTTPS safebrowsing.googleapis.com
79 2.169625900 172.16.11.157 172.16.8.1 DNS 87 Standard query 0x5f8b A safebrowsing.googleapis.com
80 2.169856500 172.16.8.1 172.16.11.157 DNS 103 Standard query response 0x5f8b A safebrowsing.googleapis.com A 142.251.222.170
Domain Name System: Protocol
Type here to search
Packets: 100 · Displayed: 4 (4.0%)
35°C


wireshark_100packets.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
arp
No. Time Source Destination Protocol Length Info
8 0.082997300 Dell_37:fa:af Broadcast ARP 60 Who has 172.16.9.180? Tell 172.16.9.184
14 0.204218100 MicroStarINT_c5:cd:... Broadcast ARP 60 Who has 172.16.11.150? Tell 172.16.10.13
18 0.320228600 HP_5c:06:39 Broadcast ARP 60 Who has 172.16.10.164? Tell 172.16.10.163
19 0.373334800 HP_5b:db:ac Broadcast ARP 60 Who has 172.16.10.174? Tell 172.16.10.144
22 0.449429500 HP_5b:dd:f0 Broadcast ARP 60 Who has 172.16.10.74? Tell 172.16.10.165
24 0.599647300 MicroStarINT_c5:c8:... Broadcast ARP 60 Who has 172.16.10.245? Tell 172.16.8.240
25 0.671855700 Sophos_cf:be:45 Broadcast ARP 60 Who has 172.16.11.212? Tell 172.16.8.1
28 0.745608700 MicroStarINT_c5:cb:... Broadcast ARP 60 Who has 172.16.9.133? Tell 172.16.10.9
29 0.764633900 EliteGroupCo_14:8a:... Broadcast ARP 60 Who has 172.16.9.34? Tell 172.16.10.199
30 0.843863400 MicroStarInt_cc:27:... Broadcast ARP 60 Who has 172.16.9.133? Tell 172.16.9.66
32 0.944818900 HP_5b:dd:f0 Broadcast ARP 60 Who has 172.16.11.5? Tell 172.16.10.165
33 0.966432100 ASUSTekCOMPU_99:99:... Broadcast ARP 60 Who has 172.16.8.28? Tell 172.16.8.171
34 1.003725900 Dell_37:fa:af Broadcast ARP 60 Who has 172.16.9.180? Tell 172.16.9.184
35 1.043919400 HP_5b:db:ac Broadcast ARP 60 Who has 172.16.10.174? Tell 172.16.10.144
36 1.188807700 EliteGroupCo_15:eb:... Broadcast ARP 60 Who has 172.16.10.164? Tell 172.16.10.191
37 1.203872600 MicroStarINT_c5:cd:... Broadcast ARP 60 Who has 172.16.11.150? Tell 172.16.10.13
39 1.272970700 MicroStarINT_c5:ce:... Broadcast ARP 60 Who has 172.16.10.60? Tell 172.16.10.53
40 1.317212200 HP_5c:06:39 Broadcast ARP 60 Who has 172.16.10.164? Tell 172.16.10.163
41 1.368523500 Dell_35:11:a6 Broadcast ARP 60 Who has 172.16.10.67? Tell 172.16.9.194
60 1.436924000 HP_5b:ce:b4 Broadcast ARP 60 Who has 172.16.10.245? Tell 172.16.10.137
71 1.442239200 HP_5b:dd:f0 Broadcast ARP 60 Who has 172.16.10.74? Tell 172.16.10.165
74 1.695838900 Sophos_cf:be:45 Broadcast ARP 60 Who has 172.16.11.212? Tell 172.16.8.1
75 1.696576900 MicroStarINT_c5:ce:... Broadcast ARP 60 Who has 172.16.9.180? Tell 172.16.10.27
80 1.723210200 HP_5c:0b:2f Broadcast ARP 60 Who has 172.16.10.153? Tell 172.16.10.154
81 1.729071500 EliteGroupCo_14:8a:... Broadcast ARP 60 Who has 172.16.9.34? Tell 172.16.10.199
90 1.851992300 EliteGroupCo_15:ec:... Broadcast ARP 60 Who has 172.16.9.168? Tell 172.16.10.205
92 1.927709100 EliteGroupCo_14:8a:... Broadcast ARP 60 Who has 172.16.9.70? Tell 172.16.10.179
93 1.984866700 HP_5b:dd:f0 Broadcast ARP 60 Who has 172.16.11.5? Tell 172.16.10.165
94 2.002119500 Dell_37:fa:af Broadcast ARP 60 Who has 172.16.9.180? Tell 172.16.9.184
0000 ff ff ff ff ff ff 58 9a 4c 37 fa
Address Resolution Protocol: Protocol
Packets: 1
Type here to search